

Relevant Formulae: Caches

Nomenclature Note

- Ratio == Rate

Cache Information

- $\text{Blocksize} = 2^{\text{\# Bits in Offset}}$
- $\text{\# Rows} = 2^{\text{\# Bits in Index}}$
- $\text{\# Entries} = \text{Cache Size (without metadata)} / \text{Blocksize}$
- $\text{\# Bits in Address} = \text{\# Bits in Tag} + \text{\# Bits in Index} + \text{\# Bits in Offset}$
- $\text{Cache Size (without metadata)} = \text{\# Rows} \cdot \text{Associativity} \cdot \text{Blocksize in bits}$
- $\text{Cache Size (with metadata)} = \text{\# Rows} \cdot \text{Associativity} \cdot (\text{Blocksize in bits} + \text{Valid Bit} + \text{\# Bits in Tag})$
- $\text{Hit Ratio} = \text{\# Hits} / (\text{\# Hits} + \text{\# Misses})$
- $\text{Miss Ratio} = 1 - \text{Hit Ratio}$
- $\text{Average Memory Access Time} = \text{Hit Time} + \text{Miss Ratio} \cdot \text{Miss Penalty}$

For Access

- $\text{BlockID} = \text{Address} // \text{Blocksize}$
- $\text{Row} = \text{BlockID} \% \text{\# Rows}$
- $\text{Tag} = \text{BlockID} // \text{\# Rows}$

Cache Access

- Hit = Same Row; Same Block
- Miss = Block empty
- Miss & Evict = Same Row; Different Block

Relevant Formulae: Virtual Memory

Information

- $\text{Page Size} = 2^{\# \text{ Bits in offset}}$
- $\text{Offset} = \text{Address} \% \text{Page Size}$
- $\text{Virtual Memory Size} = 2^{\# \text{ Bits in Virtual Address}}$
- $\text{Physical Memory Size} = 2^{\# \text{ Bits in Physical Address}}$
- $\# \text{ Virtual Pages} = \text{Virtual Memory Size} / \text{Page Size}$
- $\# \text{ Physical Pages} = \text{Physical Memory Size} / \text{Page Size}$

For Access

- $\text{Virtual Page Number} = \text{Address} // \text{Page Size}$
- $\text{Physical Page Number} = \text{Look up from the page table}$
- $\text{Physical Address} = \text{PPN} \cdot \text{Page Size} + \text{Offset}$
- $\# \text{ Bits in PPN} = \text{Physical Address Size} - \text{Offset}$
- $\# \text{ Entries} = \# \text{ Virtual Pages}$
- $\text{Page Table Entry Size} = \text{Valid Bit} + \text{Dirty Bit} + \text{Protect Bit} + \# \text{ Bits in PPN}$
- $\text{Page Table Size} = \# \text{ Entries} \cdot \text{Page Table Entry Size}$

Useful Conversions

- 1 Byte = 8 bits
- 1 MB = 2^{20} Bytes
- 1 GB = 2^{30} Bytes